

A Universal and Flux-Sensitive Entropy Lower Bound for Weakly Genuinely Nonlinear Scalar Laws

Run fa_banach_001, agent agent_lane_05

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Source problem

In arXiv:1912.00219, Capuani, Dutta, and Nguyen consider

$$u_t + f(u)_x = 0$$

on $\mathbb{R}$. For

$$\mathcal{U}_{[L,M]} = \{u_0 \in L^\infty(\mathbb{R}) : \text{supp } u_0 \subset [-L, L], \|u_0\|_\infty \leq M\},$$

they obtain an entropy upper bound for $S_T(\mathcal{U}_{[L,M]})$ when $f \in C^2$ is weakly genuinely nonlinear, meaning that f is affine on no nonempty open interval. They ask for sharp L^1 ε -entropy estimates for general fluxes in this class.

of $\mathbf{p}(E)$, $\mathbf{d}(E)$ and Ψ . In particular, if $\Psi(s) = s^\gamma$ for some $\gamma \geq 1$ and the metric space (E, ρ) is generated by a finite dimensional normed space $(\mathbb{R}^d, \|\cdot\|)$ then the ε -entropy of $\mathcal{F}_{[L,V]}^\Psi$ in $\mathbf{L}^1([0, L], \mathbb{R}^d)$ is of the order $\frac{d}{\varepsilon^\gamma}$, i.e.,

$$\mathcal{H}_\varepsilon\left(\mathcal{F}_{[L,V]}^\Psi \middle| \mathbf{L}^1([0, L], \mathbb{R}^d)\right) \approx \frac{d}{\varepsilon^\gamma}.$$

The result is applied to provide an upper estimate on the ε -entropy of a set of entropy admissible weak solutions to scalar conservation laws (1.1) with general weakly genuinely nonlinear fluxes in Theorem 3.7, which partially extends the recent one in [4]. The estimate is sharp in the case of fluxes having finite inflection points with a polynomial degeneracy. However, a natural question regarding sharp estimates of the ε -entropy for such solution sets to (1.1) with general weakly genuinely nonlinear fluxes is still open.

This paper is organised as follows. In Section 2, we present some preliminary results on covering and packing numbers of a totally bounded metric space and also include necessary concepts related to functions of bounded total generalized variation. In Section 3, the first subsection focuses on finding the upper and lower estimates of the ε -entropy for a set of bounded total generalized variation functions, while the second subsection is an application

The result below supplies a general lower estimate and a sharper flux-sensitive version. It does not identify the sharp rate for every such flux, so its status is partial.

Statement

Write $\mathcal{H}_\varepsilon(K \mid L^1)$ for the base-two logarithm of the least number of L^1 -balls of radius ε covering K .

Theorem 1 (Universal lower bound). *Let $f \in C^2(\mathbb{R})$ be weakly genuinely nonlinear. For every $L, M, T > 0$, there are constants $c > 0$ and $\varepsilon_0 > 0$, depending on f, L, M, T , such that*

$$\mathcal{H}_\varepsilon(S_T(\mathcal{U}_{[L,M]})) \mid L^1(\mathbb{R})) \geq \frac{c}{\varepsilon} \quad (0 < \varepsilon < \varepsilon_0).$$

The proof gives the following scale-sensitive strengthening. After replacing $f(u)$ by $f(u) - f'(0)u$, which only translates solutions, suppose that for a given small ε there are $a = a_\varepsilon$, $\delta = A\varepsilon/L$, and $K = K_\varepsilon > 0$ such that

1. $[a, a + \delta] \subset (-M, M)$, and f'' has one strict sign there;
2. $\max_{[a, a + \delta]} |f''| \leq K$;
3. the interval joining 0 to $[a, a + \delta]$ is contained in a fixed sufficiently small neighbourhood of zero.

Here $A > 0$ is a universal sufficiently large constant. Whenever $TK\delta$ is small relative to L , the same construction yields

$$\mathcal{H}_\varepsilon(S_T(\mathcal{U}_{[L,M]})) \mid L^1(\mathbb{R})) \geq c_0 \frac{L^2}{TK\varepsilon}, \quad (1)$$

with a universal numerical $c_0 > 0$. Thus locally flatter windows carry more independent pulses. For example, if windows of length comparable to ε satisfy $K_\varepsilon \lesssim \varepsilon^{m-1}$, then (1) gives an ε^{-m} lower bound.

Independent-pulse packing lemma

Lemma 1. *Assume $f'(0) = 0$. Let $[a, a + \delta] \subset (-M, M)$ be an interval on which f'' has one strict sign and $|f''| \leq K$. Assume also that all Riemann-wave speeds between the states 0 and $[a, a + \delta]$ have absolute value at most $L/(16T)$. If δ and $q := 8TK\delta$ are sufficiently small compared with L , then $S_T(\mathcal{U}_{[L,M]})$ contains a family $\{v_\sigma\}_{\sigma \in \mathcal{C}}$ such that*

$$|\mathcal{C}| \geq 2^{N/8}, \quad N \geq \frac{L}{16q}, \quad \|v_\sigma - v_\tau\|_1 \geq \frac{L\delta}{64} \quad (\sigma \neq \tau).$$

Proof. Place a plateau of height a on $[-3L/4, 3L/4]$, with value zero outside. Inside $[-L/4, L/4]$, divide the line into N blocks of length $4q$. In each block choose a fixed subinterval I_j of length q . For $\sigma = (\sigma_1, \dots, \sigma_N) \in \{0, 1\}^N$, define

$$u_\sigma(0, x) = a + \delta \sum_{j=1}^N \sigma_j \mathbf{1}_{I_j}(x)$$

on the plateau and set it equal to zero outside the plateau. These initial data belong to $\mathcal{U}_{[L,M]}$.

Every wave created by a pulse takes values in $[a, a + \delta]$. Its speed differs from $f'(a)$ by at most $K\delta$, by the mean-value theorem (the same bound applies to shock speeds). During time T , therefore, each edge moves relative to $f'(a)T$ by at most $TK\delta = q/8$. The pulse wave fans do not meet one another because the blocks have length $4q$. The waves generated at the two ends of the large plateau stay away from the pulse zone: its initial buffer is $L/2$, while the assumed speed bound permits closing by at most $L/8$.

Consequently, up to time T , the solution in the cone of the j -th block depends only on σ_j . If $\sigma_j = 1$, comparison with the constant solution a , together with conservation of the compactly supported mass difference, gives

$$\int_{\text{the } j\text{-th cone}} |S_T u_\sigma - a| dx = \delta q.$$

If $\sigma_j = 0$, that contribution is zero. Thus two codewords differing in $d_H(\sigma, \tau)$ coordinates satisfy

$$\|S_T u_\sigma - S_T u_\tau\|_1 \geq d_H(\sigma, \tau) \delta q. \quad (2)$$

The elementary greedy binary-code bound supplies $\mathcal{C} \subset \{0, 1\}^N$ with $|\mathcal{C}| \geq 2^{N/8}$ and minimum Hamming distance at least $N/4$ for all sufficiently large N . Since the blocks fit in an interval of length $L/2$, one may take $N = \lfloor L/(8q) \rfloor$, and for small q this is at least $L/(16q)$. Combining this with (2) proves the stated separation. $\square$

Proof of the results

Subtracting the affine term $f'(0)u$ from the flux corresponds to a common spatial translation at time T , hence preserves all L^1 distances. We may assume $f'(0) = 0$.

For the window estimate, take A large enough that $L\delta/64 > 2\varepsilon$. Lemma 1 then gives a 2ε -separated family of cardinality at least $2^{N/8}$, and therefore

$$\mathcal{H}_\varepsilon(S_T(\mathcal{U}_{[L,M]})) \mid L^1 \geq \frac{N}{8} \geq c_0 \frac{L}{TK\delta} = c'_0 \frac{L^2}{TK\varepsilon}.$$

For Theorem 1, continuity of f' at zero gives a neighbourhood $(-\eta, \eta)$ in which all characteristic and chord speeds, after the normalization, are smaller than $L/(16T)$. Weak genuine nonlinearity says that $\{u : f''(u) \neq 0\}$ is dense. Choose one $a \in (-\eta/2, \eta/2)$ with $f''(a) \neq 0$. By continuity there is a fixed interval $[a, a+r] \subset (-\eta, \eta) \cap (-M, M)$ on which f'' has one strict sign and $|f''| \leq K$ for a fixed finite K . For all sufficiently small ε , the choice $\delta = A\varepsilon/L < r$ satisfies the lemma. Formula (1), with this fixed K , proves the universal c/ε bound.

Upgrade attempts and obstruction to a full answer

Five focused routes were checked.

1. A fixed locally curved state window yields Theorem 1.
2. Allowing the window to depend on ε yields the stronger curvature-sensitive estimate (1); it recovers the usual polynomial exponents and also applies along selected scales for fluxes with infinitely many inflection points.
3. Adapting the lower construction of Ancona–Glass–Nguyen confirms that one-sided controllability gives the same mechanism when curvature keeps one sign from the zero state, but that hypothesis is absent in the target.
4. Trying to match the source’s generalized-variation upper bound fails at a precise point: weak genuine nonlinearity only makes $\{f'' \neq 0\}$ open dense. Its sign components near a flat accumulation point may all be much shorter than the required amplitude scale $\delta \asymp \varepsilon$. The independent-pulse lower construction then cannot access the flatness that controls the global convex-envelope modulus.

5. Allowing pulses to cross many curvature-sign changes does not repair this. The Riemann solution is governed by convex/concave envelopes and can erase the small oscillatory curvature; the disjoint backward-controllability argument no longer supplies a matching packing.

Thus the packet proves a general lower bound and a useful conditional refinement, but a sharp two-sided rate for every weakly genuinely nonlinear C^2 flux remains open.

Novelty and literature check

The cheap run indexes were searched for arXiv 1912.00219 and the core phrases concerning weak genuine nonlinearity, entropy, generalized variation, and scalar conservation laws; no duplicate packet or attempt was found. Bounded primary-source searches found the source paper and Ancona–Glass–Nguyen, arXiv:1806.07758, which treats strictly convex fluxes and a single polynomially degenerate inflection point. No later arXiv paper claiming the general sharp estimate, or this local-window extension, was found. The result should therefore receive both PDE-proof and novelty review.

References

- [1] R. Capuani, P. Dutta, and K. T. Nguyen, *Metric entropy for functions of bounded total generalized variation*, arXiv:1912.00219.
- [2] F. Ancona, O. Glass, and K. T. Nguyen, *On Kolmogorov entropy compactness estimates for scalar conservation laws without uniform convexity*, arXiv:1806.07758; SIAM J. Math. Anal. 51 (2019), 3020–3051.